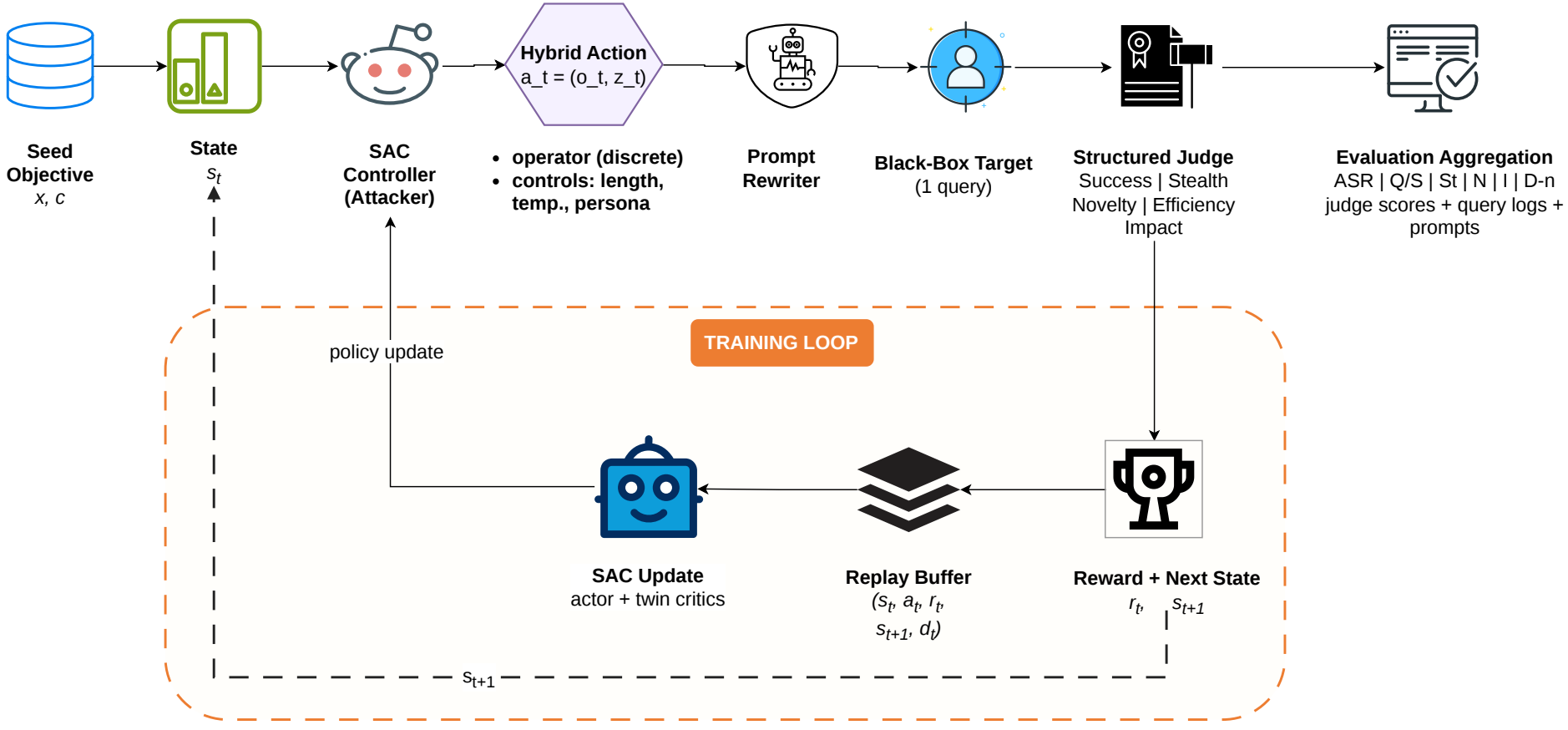

Seed Objective
x, c
State
s_t
SAC Controller (Attacker)
Hybrid Action
a_t = (o_t, z_t)
operator (discrete)
controls: length, temp., persona
Prompt Rewriter
Black-Box Target
(1 query)
Structured Judge
Success | Stealth
Novelty | Efficiency
Impact
Evaluation Aggregation
ASR | Q/S | St | N | I | D-n
judge scores + query logs + prompts
TRAINING LOOP
policy update
SAC Update
actor + twin critics
Replay Buffer
$(s_t, a_t, r_t, s_{t+1}, d_t)$
Reward + Next State
r_t s_{t+1}
s_{t+1}